

The Burger Grind — Three Cuts, Two Passes, Salt at the Pan

Chuck, brisket and short rib at 2:1:1, ground twice from near-frozen and salted only as it meets the pan — an open, tender grain no tub of mince can give. The probe decides doneness; colour never does.

Yield: **12 × 150g** (10–12) Active: **35 min** Total: **1 hr 15** Pan: **smoking** (~230°C) Done: **55–57°C** guest · **71°C** safe

WHERE IT WAS BORN

A port, not a dish. Hannah Glasse's *Hamburg sausage* (in print by 1774) was beef and suet, salted, spiced and chimney-smoked to keep a year; Fannie Farmer's 1896 *Hamburg Steak* made it fresh — chopped lean neck and round, salted before shaping.

EATEN, AUTHENTICALLY

In the hand, at once, without ceremony. White Castle put the small griddled patty in a bun at Wichita in 1921 and sold it by the sack — the smash is the older American form, the thick pub patty the later one.

HOW IT'S USED NOW

One grind, two ways: thick and rosy for the plate, thin and laced for the griddle. Since the 1993 outbreak the safe number is 71°C; the chef's number is 55–57 — know which one you are serving.

INGREDIENTS

WEIGHT (LB/OZ) · CUPS / VOLUME IN RIGHT COLUMN

THE GRIND — 4 LB (BUY 4½ LB)

Beef chuck (<i>boneless, fat on</i>)	1 lb	15.7 oz	
Beef brisket (<i>point end, 1 cm cap</i>)		15.9 oz	
Beef short rib (<i>boned</i>)		15.9 oz	
AT THE PAN			
Fine sea salt (<i>crust only, never in the mix</i>)	0.85 oz	4 tsp	
Black pepper, coarse	0.21 oz	2½ tsp	
Beef dripping (<i>or neutral oil</i>)	1.1 oz	2 tbsp	
Soft buns (<i>potato or brioche</i>)	12		

THE RATIO — THE BLEND = 100%
Chuck 50% · Brisket 25% · Short rib 25% → fat 20–25% · Salt 0% inside, 1.3% on the crust · Plates 10 mm then 4.5 mm. Twenty per cent fat is the knee of the curve: cooking loss stays flat to 20%, then climbs (Troutt 1992). 150g a patty for the plate; 2 × 75g for the smash.

Kit: grinder with 10 mm + 4.5 mm plates (blade, auger, bowl all frozen) · scale · two trays · bowl over ice · heavy pan or plancha · stiff spatula · probe thermometer · greaseproof

METHOD

- Dice and chill everything.** 2–3 cm dice, gristle and silverskin out, fat in. Tray, freezer **25–30 min**: edges firm and frosted, centre still gives. Grinder parts and bowl in the freezer too.

Warm fat smears grey over the lean; near-frozen fat shears into chips that survive as juice pockets. Raw mince lives at 4°C or below. — López-Alt; USDA FSIS
- First grind, coarse.** 10 mm plate, dice fed steadily, never forced, into the cold bowl over ice; the three cuts go through mixed. Grey smear at the plate = warm: stop, freeze 10 min.
- Re-chill, second grind, finer.** 10 min in the freezer, then the **4.5 mm** plate (6 mm for a looser pub patty), once more over ice. Do not stir; the fat must still show as white flecks.

The first pass opens the dice; the second sets the grain and gives a thick patty its bind — without salt.
- Portion, dimple — no salt.** 150g, gathered loosely, pressed to **2 cm** and a thumb wider than the bun; dimple the centre. Tray, cover, fridge **30 min** (up to 24 h).

Salt swells the myofibrils and extracts myosin, the glue of a sausage: salt worked in gives a springy patty, salt on the surface keeps it loose. — Offer & Trinick 1983; Ruusunen & Puolanne 2005
- Sear hard, salt at the pan.** Dripping just smoking (~230°C). Patties in, not touching; salt the top face now (~1g) with the pepper. **2–3 min** until the crust releases; flip once, salt the second face. Never press.

A wet face is pinned at 100°C by its own steam until it dries; browning only starts after. — Martins et al. 2000
- Probe, pull, rest.** From the side into the centre. Safe for any ground beef: **71°C** (USDA). Chef's medium-rare: pull at **52–54°C**, rest 3 min to **55–57°C** — an informed risk for guests who choose it, never for children, pregnancy, the elderly or the immunocompromised. Grinding folds surface bacteria through the mass; a fresh, cold, whole-cut grind lowers the risk and never removes it. Buns in the pan fat; build.

1992–93: 500+ *E. coli* O157:H7 cases and four deaths traced to one chain's patties; the FDA's interim line went to 155°F, USDA's now sits at 71°C. — CDC MMWR 42(14) 1993

No grinder aboard — the hand-chop

- a. Freeze the dice 30 min; slice 3 mm, stack, cross-cut.
- b. Rock-chop with two knives to 3–5 mm, folding in from the edges; straight back to the fridge.
- c. Portion as above — it holds better as a smash than as a thick patty.

Done: the probe, not the colour — patties can be brown right through at 55°C and the juices never run clear (Hague 1994). Guests 55–57°C after the rest; the vulnerable and the smash 71°C. Crust dark and releasing on its own.
Serve: rest 3 min; buns in the pan fat; hold cooked patties no more than 5 min.

ONE GRIND, TWO DISHES — THE CREW SMASH

Same 1.8 kg = 12 doubles. **2 × 75g** loose balls a bun on a plancha as hot as it goes: ball down 10 s, **smash through greaseproof to 4–5 mm**, once; salt now; 60–90 s until the edge laces; scrape, flip, cheese, 30 s, stack. *Safe by geometry: 5 mm passes 71°C before the crust has finished — the crew, the children and the cautious all get the good version.*

ELEVATION		
TIER 1 — NO EXTRA TIME - Buns in the dripping — cut face down, 30 s - Smoked flaky salt — on the crust, at the pan - Onions into the smash — paper-thin, on the ball	TIER 2 — WORTH IT - Dry-aged trim — 10–15% of the blend - Butter-baste — after the flip, last minute 24 h pickles — cucumber, vinegar, dill	TIER 3 — SERVICE - Bone marrow — 5%, frozen, with the second grind - Beef-fat mayonnaise — the rendered dripping, warm - Two tiers — guest patty at 55°C, crew smash after

TROUBLESHOOT

Dense, bouncy, sausage-like	Salt in the mix, or over-worked → salt only at the pan; handle less
Grey, greasy, weeping	Warm meat or grinder → near-frozen dice, frozen kit, ice under the bowl
Falls apart on the flip	One coarse pass only → second grind at 4.5 mm, colder patty; or smash it
Domed puck	No dimple, or pressed → dimple the centre, hands off
Pale, no crust	Cool pan, crowded, wet surface → smoking pan, space, loose cold grind
Brown but under temp	Premature browning → the probe decides, every patty

CHARTER PREP & STORAGE

Best route	Grind on a calm morning at anchor; portion; cook within 24 h at ≤4°C
Freeze	Raw, flat between greaseproof, bagged, 1 month; thaw in the fridge overnight
From frozen	Only as a smash — a thick patty chars before the centre moves
In a swell	Clamp the grinder; bowl in a bigger bowl of ice; tray with a lip
Never	Salt ahead, refreeze, taste the raw grind, grind warm meat